

LIFEIFY — COMPLETE PROJECT DOCUMENTATION SET (WEB APP)

> ## ⚠️ ONE FINAL PROFESSIONAL WARNING — THEN I BUILD WHAT YOU ASKED

>

> You've chosen to include **diagnostic symptom analysis**, **AI medication verdicts**, and **clinical mental health programs**. I'm building the docs with these included as instructed — but with **mandatory guardrails baked into every document** (disclaimers, red-flag routing, clinical data sources, and an AI Safety Layer as a required component). Without these, your app is a legal and ethical time bomb; with them, it's a defensible product direction. Please also budget for a **medical advisor and a lawyer before launch** — this is a requirement, not a suggestion, given these three features.

>

> **Web app adaptation note:** Since this is a **web app (not native mobile)**, phone-sensor step tracking is not reliably available in browsers. The plan below adapts: activity tracking becomes manual/PWA-based where possible, and the core value (AI + reports + trends) is fully web-viable. A PWA (installable web app) gives you app-like behavior.

📄 DOCUMENT 1 — PRD (Product Requirements Document)

App Name & Tagline

LIFEIFY — "Don't Just Track Your Health. Understand It."

The Problem

Health data is fragmented across apps and paper. Lab reports are unreadable. Medication effects go untracked. Mental health is ignored. Nobody explains what it all means. Adults 18–50 in India want one place that understands their whole health picture.

Core Value Proposition

An AI-powered web platform that unifies fitness, nutrition, sleep, medical records, medications, and mental health into one dashboard — and uses AI to **analyze, explain, and judge** what the data means, including symptom diagnosis assistance, medication effectiveness verdicts, and structured mental health programs.

Target User Persona

Riya, 29, a marketing professional in Pune. She has slightly elevated blood pressure, takes one medication, gets annual blood tests she never understands, feels stressed but has never addressed it, and wants one place that tells her what's actually going on with her body — without buying a smartwatch.

Feature List

MUST HAVE (MVP)

1. Email/OTP signup + login
2. Onboarding quiz (goals, conditions, medications, family history, mental state)
3. Health Score dashboard (0–100)
4. Health metrics logging: food (Indian DB ~2,000 items), sleep, water, mood, weight, BP, blood sugar
5. ****AI Health Assistant chat**** (context-aware, 5 free questions/month)
6. ****AI Symptom Analysis**** — user describes symptoms → AI returns structured analysis with possible causes, severity triage, and recommended action (with mandatory disclaimers)
7. Medical record upload → OCR → AI plain-language report analysis (1 free/month)
8. ****Medication Progress View**** with ****AI effectiveness verdict**** ("your data suggests this medication is likely/not likely achieving its intended effect — discuss with your doctor") — includes vitals trends, adherence, side-effect check-ins
9. Gamified weekly Med Check-In (emoji slider, side-effect cards, badges, streaks)
10. ****Mental Health Program**** — structured 4–8 week guided modules (mood tracking, CBT-style exercises, breathing sessions, progress scoring) — NOT therapy; clearly labeled self-guided wellness
11. Mood/emotion daily check-in with streaks
12. 30-day trends & charts
13. Shareable doctor summary (PDF)
14. Notifications (in-app + email)
15. Free plan + Premium (₹199/mo, ₹1,499/yr) + Razorpay UPI AutoPay
16. Usage limits + contextual upgrade prompts
17. Data export/delete
18. Admin dashboard
19. ****AI Safety Layer**** (red-flag detection, emergency routing, disclaimers, audit log)

NICE TO HAVE (V1)

- Family profiles + caregiver alerts
- Voice logging
- Health Connect / Apple Health import
- Hindi language
- Advanced analytics & correlations
- Anomaly alerts

User Stories

- As a **user**, I want to ask the AI "why am I always tired?" so that I get an answer based on MY data.
- As a **user**, I want to upload my blood test report so that it's explained in plain language.
- As a **user with BP**, I want to log my medication and readings so the AI tells me whether my treatment is working.
- As a **stressed user**, I want a guided daily mental health program so I can improve without therapy costs.
- As a **user**, I want one Health Score so I can see if I'm improving overall.
- As a **free user**, I want to try the AI so I can decide whether to pay.

Out of Scope (this version)

- Doctor consults/telemedicine marketplace
- Emergency SOS service
- Wearable hardware
- Human coaches
- Native mobile apps (web/PWA only for v1)
- Drug-drug interaction AI warnings (licensed DB integration is V1)

Success Metrics

- Activation: % of signups completing quiz + first AI question + one log within 48h
- Time to first value: < 5 minutes
- D7/D30 retention
- AI questions per active user
- Free → Premium conversion rate
- MRR, churn
- **Safety metric: % of AI answers reviewed by safety layer; zero unresolved red-flag incidents**

📄 DOCUMENT 2 — TRD (Technical Requirements Document)

Stack

| Layer | Choice | Why |

|---|---|---|

| Frontend | **Next.js 14+ (React, TypeScript)** | SSR for marketing + app pages, one codebase, PWA support |

| UI library | Tailwind CSS + shadcn/ui | Fast, consistent components |

| Backend | **Python FastAPI** | Best AI/LLM tooling, your stated preference |

| Database | **PostgreSQL on Supabase** (or Neon) | Managed, RLS for health data, auth included |

| Auth | **Supabase Auth** (email + OTP) | Simple, built-in RLS integration |

| AI | OpenAI / Anthropic API (configurable) | LLM with context-engine prompting; medical safety layer wraps all calls |

| OCR | Google Cloud Vision API | Lab report text extraction (English + Hindi) |

| Payments | **Razorpay** (UPI AutoPay + cards) | India-first subscriptions |

| Email | Resend | Transactional + weekly digests |

| File storage | Supabase Storage (S3-backed) | Report PDFs/images, encrypted |

| Hosting | Frontend: **Vercel** · Backend: **Railway** (or Fly.io) | Standard, low-ops |

| Analytics | PostHog | Product events |

| Errors | Sentry | Both frontend & backend |

| PDF generation | React-PDF / server-side reportlab | Doctor summaries |

Folder Structure

```

...
lifify/
├── apps/
│   ├── web/                # Next.js frontend
│   │   ├── app/
│   │   │   ├── (marketing)/ # Public pages: /, /pricing, /privacy, /terms
│   │   │   ├── (auth)/      # /login, /signup
│   │   │   ├── (app)/       # Protected app pages
│   │   │   ├── dashboard/
│   │   │   ├── assistant/
│   │   │   ├── track/
│   │   │   ├── records/
│   │   │   ├── medications/
│   │   │   ├── mental-health/
│   │   │   ├── insights/
│   │   │   └── settings/
│   │   └── admin/          # Admin dashboard (role-gated)
│   ├── components/        # Reusable UI components
│   ├── lib/               # API client, helpers
│   ├── public/            # PWA manifest, icons
│   └── api/               # FastAPI backend
│       ├── app/
│       │   ├── routers/    # auth, health_logs, ai, records, meds,
│       │   │               # mental_health, billing, admin
│       │   ├── services/   # ai_engine, safety_layer, ocr, insights
│       │   ├── models/     # SQLAlchemy ORM models
│       │   ├── core/       # config, security, dependencies
│       │   ├── prompts/    # LLM prompt templates (versioned)
│       │   └── tests/

```

Naming Conventions

- Files: `kebab-case` · React components: `PascalCase` · Python: `snake\_case`
- DB tables: `snake\_case`, plural (`health\_logs`)
- API routes: `/api/v1/<resource>` (versioned from day one)

Environment Variables

```

...

DATABASE_URL, SUPABASE_URL, SUPABASE_ANON_KEY, SUPABASE_SERVICE_KEY,
SUPABASE_JWT_SECRET, OPENAI_API_KEY (or ANTHROPIC_API_KEY),
AI_MODEL_NAME,
GOOGLE_VISION_API_KEY, RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET,
RAZORPAY_WEBHOOK_SECRET, RESEND_API_KEY, SENTRY_DSN, POSTHOG_KEY,
APP_URL, ENVIRONMENT, ADMIN_EMAILS

```

...

Hard Constraints

- **All AI medical outputs must pass through the safety layer** — no exceptions, enforced at the router level, not in the frontend.
- All health data encrypted at rest; PII fields encrypted at application level.
- All LLM prompts version-controlled and reviewed before deploy.
- Rate limits on AI endpoints (cost control + free-tier limits).
- PWA enabled (manifest + service worker) for install/mobile-like use.

DOCUMENT 3 — APP FLOW (Navigation & User Journey Map)

Screens / Pages

Screen	Description
`/`	Landing Hero, demo (report explained in 30s), features, free-vs-paid, pricing, FAQ, CTA
`/signup` , ` /login`	Email/OTP auth
`/onboarding`	3-step quiz: basics → health/meds → goals & mental state
`/dashboard`	Health Score ring, metric cards, AI Insight of Day, streak, quick-log
`/assistant`	AI chat (health Q&A, symptom analysis, med verdicts)
`/track`	Tabs: Food / Sleep / Vitals / Mood / Water / Weight
`/records`	Timeline of uploads, OCR results, AI summaries, doctor summary generator
`/medications`	Med list, reminders, Progress View (verdicts), Med Check-In game
`/mental-health`	Current program, daily module, mood timeline, streaks
`/insights`	Trends, charts, correlations, anomaly alerts
`/settings`	Profile, privacy, export/delete, billing, plan
`/admin`	Users, subscriptions, refunds, usage, AI audit log, announcements
`/pricing` , ` /privacy` , ` /terms` , ` /help`	Public pages

Navigation Structure

- **Marketing pages:** top navbar + footer.
- **App:** **left sidebar on desktop** (Dashboard, Assistant, Track, Records, Medications, Mental Health, Insights, Settings) with **bottom tab bar on mobile** (Dashboard, Assistant, Track, More). Persistent **"+" quick-log** button on all app screens.

Entry Points & Auth Flow

- Visitor → `/` → CTA → `/signup`
- `/signup` (email) → OTP verify → `/onboarding` (3 steps) → `/dashboard`
- Returning user → `/login` → OTP → `/dashboard` (redirect if onboarding incomplete)
- Logged-out access to protected route → `/login?redirect=<path>`

Key User Journeys

****Journey 1 — First value (report explanation):****

Dashboard (empty-state prompt) → Records → Upload report → "Analyzing..." state → OCR confirm values → AI analysis rendered (plain language + flagged values + disclaimers) → "Ask the AI about this report" → Assistant

****Journey 2 — Symptom analysis:****

Assistant → "I've had headaches for 3 days" → AI asks 2–3 follow-ups (duration, severity, other symptoms) → Structured result card: possible causes, severity triage (informational), red-flag check, "discuss with doctor" + doctor summary option

****Journey 3 — Medication verdict:****

Medications → Add med (name, dose, condition) → Log readings + weekly Med Check-In game → after 2 weeks of data → "Verdict ready" card: data summary + AI assessment ("readings suggest treatment goal is/isn't being met — your doctor makes the final call") + adherence overlay + share with doctor

****Journey 4 — Mental health program:****

Mental Health → Choose goal (stress/sleep/mood) → program starts (Day 1 module: 5-min exercise) → daily completion → streak + progress score → weekly AI recap

Edge Cases

- ****Empty states:**** every screen teaches + collects data ("No medications yet — add one to start tracking")
- ****Loading states:**** skeleton screens; OCR shows progress ("Reading your report... 80%")
- ****Error states:**** AI unavailable → "Our AI is resting — try again in a minute" (never leave user with blank failure); OCR fails → "Report image too blurry — retake?"
- ****Limit states:**** AI question 5/5 used → upgrade card (value-stated, once)
- ****Safety interlock:**** any red-flag input (e.g., "chest pain") → immediate crisis card, chat disabled for that thread, incident logged
- ****Modals:**** quick-log bottom sheet, med-add modal, upgrade modal (max 1/week), record value confirmation, crisis intervention card

Redirect Logic

- Onboarding incomplete → always back to onboarding
- Free user hits limit → inline upgrade prompt (not a forced redirect)
- Payment success → back to the feature that triggered the upgrade, unlocked
- Admin role users → `/admin` link visible in settings

DOCUMENT 4 — UI/UX DESIGN BRIEF

Aesthetic Direction

"Calm intelligence." Friendly, modern, trustworthy — positioned between fitness-app energy and hospital coldness. Generous whitespace, insight cards that feel like a smart friend's message.

Color Palette

| Token | Value | Use |

|---|---|---|

| Background | `#FAF8F8` (warm off-white) | Base |

| Surface | `#FFFFFF` | Cards |

| Primary text | `#1A2B33` (deep slate) | Headlines/body |

| Secondary text | `#5E7A85` | Captions, labels |

| **Primary/CTA** | `#0E7C6B` (calm teal) | Buttons, links, active states |

| Accent | `#F4A259` (warm amber) | Streaks, badges, highlights |

| Success | `#3B9B6F` | Positive readings, confirmations |

| Warning | `#E5A33D` | Trend alerts, flagged values |

| Error/Crisis | `#C4453C` | Errors, red-flag cards (used sparingly, high contrast) |

Typography

- **Headings:** Inter (or Plus Jakarta Sans) — Bold, 28/22/18px hierarchy

- **Body:** Inter Regular, 16px, 1.6 line-height

- **Captions/labels:** 13px, Medium

- Chat/AI answers: 15–16px, max ~70 characters per line for readability

Component Style

- Rounded corners: 12px cards, 8px buttons/inputs

- Flat design with subtle shadows (`0 2px 8px rgba(0,0,0,0.06)`)

- Pill-shaped chips (suggested questions, med badges)

- Cards, not tables, for health data; tables only in admin

- Emoji-integrated by design (check-in game, mood) — this is a deliberate warmth signal

- Dark mode: **Phase 2** (light first, tokens structured to allow it)

Key UI Patterns

- Insight Card (AI output standard: insight → why → next step → disclaimer)

- Bottom-sheet modals for quick logging

- Streak flames & badge shelf (gamification)

- Chat interface with typing indicator and structured answer cards (not walls of text)

- Progress ring (Health Score)

- Chart cards with 7/30/90-day range toggles

Mobile Responsiveness

- Sidebar → bottom tabs under 768px

- All logging flows ≤ 2 taps

- Charts legible at 375px width

- Sticky "+" button, thumb-reachable

- PWA installable

Accessibility

- WCAG AA contrast (the teal/white pairing passes; verify all tokens)
- 16px minimum body text
- All interactive elements $\geq$ 44px tap targets
- Emoji always paired with text labels (screen readers + clarity)
- Color never the sole indicator for warnings (icon + text always)

Inspiration

Calm's tranquility + Notion's clarity + HealthifyMe's engagement loops — without the aggressive upselling.

📄 DOCUMENT 5 — BACKEND SCHEMA (Data Model & Auth)

Database Tables (PostgreSQL)

```sql

users (id uuid PK, email unique, phone, name, dob, sex, height\_cm, weight\_kg, role enum('user','admin'), onboarding\_complete bool, is\_premium bool, plan enum('free','premium'), created\_at)

health\_profiles (id PK, user\_id FK→users, goals jsonb, conditions jsonb, family\_history jsonb, diet\_type, activity\_level, updated\_at)

health\_logs (id PK, user\_id FK, log\_type enum('food','sleep','water','mood','weight','bp','blood\_sugar','steps'), value numeric, unit, metadata jsonb, logged\_at) -- INDEX: (user\_id, log\_type, logged\_at)

foods (id PK, name, name\_hi, calories, protein, carbs, fat, serving\_g, category) -- INDEX: name trgm (search)

medications (id PK, user\_id FK, name, dose, frequency, condition\_treated, start\_date, end\_date, active bool)

med\_checkins (id PK, medication\_id FK, week\_number, feeling\_score int, side\_effects jsonb, comparison enum('worse','same','better'), adherence int, created\_at) -- gamified check-in data

med\_verdicts (id PK, medication\_id FK, generated\_at, data\_summary jsonb, ai\_assessment text, confidence enum, disclaimer\_shown bool, doctor\_acknowledged bool, safety\_reviewed bool)



-- every AI verdict row is auditable

medical\_records (id PK, user\_id FK, file\_url, file\_type, uploaded\_at,  
ocr\_raw\_text text, extracted\_values jsonb, ai\_summary text,  
analysis\_count int)

ai\_conversations (id PK, user\_id FK, created\_at)

ai\_messages (id PK, conversation\_id FK, role enum('user','assistant'),  
content text, safety\_flags jsonb, red\_flagged bool, created\_at)  
-- INDEX: (conversation\_id)

symptom\_analyses (id PK, user\_id FK, symptoms jsonb, followups jsonb,  
ai\_result jsonb (possible\_causes, triage, recommended\_action),  
red\_flag bool, created\_at)

mental\_health\_programs (id PK, user\_id FK, program\_type enum, start\_date,  
current\_day int, completed\_days jsonb, progress\_score int)

mh\_daily\_entries (id PK, program\_id FK, day int, module\_id, completed bool,  
mood\_before int, mood\_after int, notes)

health\_scores (id PK, user\_id FK, score int, components jsonb, calculated\_at)  
-- INDEX: (user\_id, calculated\_at)

reminders (id PK, user\_id FK, type, schedule jsonb, active bool)

subscriptions (id PK, user\_id FK, razorpay\_sub\_id, plan enum, status enum,  
current\_period\_end, created\_at)

payments (id PK, subscription\_id FK, amount, status, razorpay\_payment\_id)

usage\_counters (id PK, user\_id FK, month date, ai\_questions int,  
report\_analyses int, storage\_mb numeric) -- INDEX: (user\_id, month)

usage\_events (id PK, user\_id FK, event\_type, metadata jsonb, created\_at)  
-- analytics: activation, funnel

ai\_safety\_log (id PK, user\_id FK, trigger text, classification,  
action\_taken, resolved bool, created\_at) -- audit trail

family\_members (id PK, owner\_user\_id FK, name, relationship, access\_level)  
...

## Relationships

`users` 1→many: health\_logs, medications, medical\_records, ai\_conversations, subscriptions, usage\_counters. `medications` 1→many: med\_checkins, med\_verdicts.  
`mental\_health\_programs` 1→many: mh\_daily\_entries.

### ## Auth & Access Model

- **\*\*Supabase Auth + Row Level Security (RLS)\*\*** on every health table: `user\_id = auth.uid()` — a user can only ever read/write their own health rows.
- Admin role: service-role access via `ADMIN\_EMAILS` + separate `/admin` routes, server-side gated.
- Caregiver/family access (V1): explicit permission rows only.
- **\*\*Sensitive fields encrypted at app level:\*\*** OCR raw text, AI conversations, symptom analyses (application-layer encryption key separate from DB).

### ## Storage Structure

`supabase-storage/records/{user\_id}/{record\_id}.{ext}` — private bucket, signed URLs only, 15-min expiry.

### ## Event Triggers / Webhooks

- Razorpay webhook → subscription activation/renewal/failure
- `ai\_messages` insert → safety-layer classification job
- Daily cron → Health Score recalculation, insight generation, reminder dispatch, weekly digest email
- `usage\_counters` threshold → in-app limit flag set

### ## API Endpoints (v1)

...

POST /api/v1/auth/otp/send, /auth/otp/verify

POST /api/v1/onboarding

GET /api/v1/dashboard

POST /api/v1/logs · GET /api/v1/logs?type&range

GET/POST /api/v1/foods/search?q=

POST /api/v1/ai/chat (rate-limited, safety-wrapped)

POST /api/v1/ai/symptom-analysis (safety-wrapped)

POST /api/v1/medications · POST /api/v1/medications/{id}/checkin

GET /api/v1/medications/{id}/verdict (safety-wrapped, generates on demand)

POST /api/v1/records/upload · POST /api/v1/records/{id}/analyze

GET /api/v1/records/{id}/doctor-summary.pdf

POST /api/v1/mental-health/start · POST /api/v1/mental-health/day/{n}/complete

GET /api/v1/insights/trends

POST /api/v1/billing/checkout · POST /api/v1/webhooks/razorpay

GET /api/v1/account/export · DELETE /api/v1/account

Admin: /api/v1/admin/users, /admin/subscriptions, /admin/safety-log

...

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## # 📄 DOCUMENT 6 — IMPLEMENTATION PLAN

### #### PHASE 1 — Project Setup ✅ Done when:

Repo created (monorepo: `web` + `api`), folder structure per TRD, env vars templated (`.env.example`), Supabase project created, Vercel + Railway accounts connected, CI (GitHub Actions: lint + test on PR), Sentry + PostHog wired.

### #### PHASE 2 — Database & Migrations ✅ Done when:

All tables in Doc 5 migrated, RLS policies written and **tested with two users (each cannot see the other's data)**, storage bucket + signed URL flow works, seed script loads 2,000-item food database.

### #### PHASE 3 — Authentication ✅ Done when:

Signup → OTP → session works; onboarding quiz saves profile; logged-out protection + redirects work; passwordless email magic link as fallback; admin role gating works.

### #### PHASE 4 — Core Features (in dependency order) ✅ Done when:

1. **Logging engine** (health\_logs + food search + dashboard cards render real data)
2. **AI Assistant + Safety Layer** — chat works with user context injected; **safety layer blocks red-flags** (tested with 50 adversarial prompts); every answer carries disclaimer; audit log populated
3. **Records: upload → OCR → confirm → AI analysis** end-to-end on real lab reports (test 20 varied Indian lab formats)
4. **Symptom analysis flow** with follow-ups + structured result + red-flag routing
5. **Medications: tracking → check-in game → verdict generation** (with doctor-final-call framing + audit row)
6. **Mental health: program engine + 2 starter programs** (stress, sleep) + daily flow + streaks
7. **Health Score + daily insight generation** (cron)
8. **Monetization: Razorpay checkout, UPI AutoPay, webhook-driven subscription state, usage counters + limits + upgrade prompts**
9. **Doctor summary PDF**
10. **Admin dashboard** (users, subs, refunds, safety log)

### #### PHASE 5 — UI Polish & Responsive ✅ Done when:

Design system implemented per Doc 4; all screens pass 375px mobile test; empty/loading/error states on every screen; PWA installable; WCAG AA verified.

### #### PHASE 6 — Testing, Errors, Edge Cases ✅ Done when:

- Unit tests on logging, limits, billing logic
- **AI safety test suite: 200+ adversarial prompts** (medical emergencies, self-harm indicators, medication misuse) — all correctly routed, zero unsafe outputs
- Razorpay webhook failure/retry tested

- OCR bad-photo fallback tested
- Usage limit boundaries tested (question 5, 6th blocked with correct message)
- Load test on AI endpoints (rate limiting works)

#### PHASE 7 — Deployment & Launch  Done when:

Production envs configured; domains + SSL live; Sentry alerts to your phone; backups daily + **\*\*restore tested once\*\***; legal pages live (privacy, terms, disclaimers — professionally drafted); Razorpay live keys; monitoring dashboard green; **\*\*medical advisor has reviewed AI outputs and disclaimer language; legal review complete.\*\***

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##  Build order for your first week (if you're vibe-coding this)

1. Phase 1 setup → 2. `users` + `health\_profiles` + auth → 3. dashboard shell with mock data → 4. logging engine → 5. AI chat (start with a good prompt + OpenAI, add the safety layer immediately after) → then follow Phase 4's order.

One last time, as your strategist: **\*\*Phases 6 and 7's safety/legal items are not optional extras\*\*** — given the three features you've chosen, they're the difference between a product and a lawsuit. Good luck, founder. 